



# Flexural retrofitting of RC buildings using GFRP/CFRP – A comparative study

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## ABSTRACT

The effectiveness of fibre reinforced polymers (FRPs) in retrofitting/repairing of the reinforced concrete (RC) components has been studied in the past to great detail. However, the seismic performance of RC structures retrofitted using FRP composites is yet to be scrutinised in terms of lateral resistance, ductility, and failure mechanism. This is of high importance if the retrofitted structures are to withstand higher seismic ground motions than they were designed for and/or pulse-type ground motions. In a comparative study, this paper reports on the results of an investigation into the flexural strengthening of RC buildings using glass/carbon fibre reinforced polymers (GFRP/CFRP). An 8-storey code-compliant RC building was considered as the case study to represent the medium-rise structures. With a slight intervention in the lateral displacement ductility and provision of the weak-beam strong-column design philosophy, the strengthening design strategy is aimed at increasing the lateral resistance. For this purpose, composite sheets are designed to be applied at the two end regions of all beams and columns on a practical flange-bonded scheme. The nonlinear pushover analysis with lumped plasticity approach was implemented in order to compare the seismic response of the original structure with the GFRP/CFRP retrofitted structures. Following validation of the adopted models, the force–deformation curves of the nonlinear plastic hinges are determined in a rigorous approach considering the material inelastic behaviour, reinforcement details, and dimensions of the members. While the nonlinear results confirm a significant increase in the lateral load carrying capacity using both composite materials, the CFRP improvement was as much as twice of the GFRP. However, the latter provides higher ductility.

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## 1. Introduction

Failure of modern engineered structures during the 1994 Northridge and 1995 Kobe earthquakes revealed the inadequacy of current design methods and code regulations for near-fault pulse-type ground motions. Despite many studies undertaken in recent years, conventional seismic design codes have not yet adopted an appropriate, accurate, and adequately simple design method, which considers the effect of velocity pulses observed in the near-fault recodes. The scarcity of reliable data and sophisticated features of the ground motions in the proximity of fault rupture are accounted as the major contributors to this issue. Furthermore, changes in the seismic hazard levels, applied loads, design methods, and serviceability requirements are amongst other reasons for retrofitting a code-compliant structure subjected to an ordinary earthquake. As a result, many existing buildings have yet to be retrofitted in order to remain reasonably safe during pulse-type ground motions or more severe earthquakes than those they have been designed for. Appropriate strengthening approaches shall be

found that allow retrofitting of inadequate buildings at a fraction of the cost of demolition and re-construction scenarios.

Amongst different methods suggested for repairing/upgrading of RC buildings, the use of externally bonded fibre reinforced polymers (FRPs) has increased significantly, especially in recent years. The inherent advantages of FRPs pose them as a more reliable candidate for seismic retrofitting of RC buildings in comparison to the traditional methods such as external bracing or steel jacketing. These include high tensile strength, low specific weight, high resistance to corrosion, and ease of application.

The efficiency of FRP retrofitting in strengthening/repairing of structural beam–column joints has been confirmed in many studies [1–5] worldwide. Researchers have also investigated the related problems such as FRP–concrete interface interaction [6–11] and creep behaviour in FRP strengthened structural members [12]. However, very few studies have scrutinised the overall behaviour of FRP-rehabilitated RC structures. Balsamo et al. [13] assessed the seismic performance of a full-scale RC structure repaired with carbon FRP (CFRP) laminates and wraps. Their experimental results proved the existence of a large displacement capacity in the repaired structure without any reduction of strength after the application of FRP at the beam–column joints and walls. In addition, the energy dissipation remains almost identical to the original structure. On the contrary, a reduction in the deformability of shear

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walls was observed during the experiments due to the presence of CFRP laminates over the entire height.

In another experimental study, Di Ludovico et al. [14] investigated seismic retrofitting of an under-designed, full-scale RC structure with FRP wrapping. In their study, a bi-directional test with peak ground acceleration (PGA) equal to 0.2 g was applied to the original structure prior to retrofitting under which the structure was found inadequate. The structure was then retrofitted in order to withstand a 50% higher PGA of 0.3 g. The successful outcome of the tests proved the effectiveness of FRP in improving the global performance of the structure in terms of ductility and energy dissipating capacity. Improving the seismic behaviour of deficient RC structures with FRP composites has also been confirmed by Garcia et al. [15] who through experimental tests and numerical modelling found that FRP retrofit results in substantial improvement of seismic performance of damaged RC frames.

Following the main trend of the argument, the current study was conducted to investigate the seismic behaviour of FRP retrofitted RC buildings. To pursue this objective, FRP sheets were applied at the beams' and columns' regions that are prone to the development of plastic hinges in such a way that increases the flexural strength. As a result, FRP sheets were applied at the top and bottom flanges of members with fibres oriented parallel to the longitudinal steel reinforcements. Of particular interest was to compare the effects of GFRP and CFRP application. As the case-study, an 8-storey moment resisting RC building was selected representing the mid-rise buildings. The seismic behaviour of the structure was evaluated using the nonlinear pushover method. In addition, the concept of lumped plasticity with flexural hinges at both ends of beams and columns was implemented in the characterisation of nonlinear properties of the structural members. The analyses were carried out in SAP 2000 [16], a commonly used finite element program by the structural engineering profession.

## 2. Research significant

Located in the Alpine–Himalayan earthquake belt, Iran has experienced devastating earthquakes. The Tabas earthquake of 1978, Manjil earthquake in 1990, Bam earthquake of 2003 and Dahoeieh–Zarand earthquake in 2005 are only the more recent examples of the many ground motions that have occurred in the region in recent decades. Most of the big cities of Iran are located in the regions near the fault rupture. Furthermore, many modern existing buildings worldwide need to be retrofitted due to the changes of the seismic hazards and applied loads. Although numerous studies have been carried out on improving the performance and load carrying capacity of RC members, mostly under seismic loads, the overall behaviour of FRP retrofitted reinforced concrete structures still needs to be elaborated. It should be pointed out that the well-known design philosophy of weak-beam strong-column is not yet adopted for moderate ductility in some structural design codes such as ACI 318 [17,18].

## 3. Design and description of the original frame

The structure considered in this study was an 8-storey moment resisting RC building. The torsional effects attributed to seismic loads have been neglected in this study and an interior planar frame was analysed as being representative of regular RC buildings. The frame considered to be part of the lateral resisting system of a residential building with three bays (each equal to 5 m). The height was assumed to be equal to 3 m for all stories. Structural design of the building complied with the intermediate (moderate) seismic provisions of ACI 318-02 [17], whereas the seismic loads were determined in accordance with the provisions of Iranian seismic

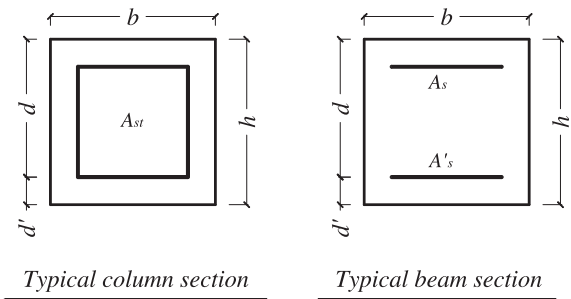


Fig. 1. Distribution of longitudinal reinforcement in a typical beam and column section.

code [19] which is similar to UBC 1994 [20]. Design base shear was calculated considering a peak ground acceleration of 0.3 g representing a high seismic hazard and soil type-III which is similar to class D of FEMA-356 [21]. In designing the moment resisting frame, the design dead load and live load were assumed to be equal to 30 kN/m and 10 kN/m, respectively. The assumed distributed gravity loads were applied to the beams in addition to the self-weight of the structure. In addition, the compressive strength of concrete was taken as 25 MPa and deformed bars of Grade 60 ( $f_y = 420$  MPa) were chosen for the steel reinforcement.

Modelling and analysis of the structure was performed in SAP 2000 [16]. The fundamental period of the structure was calculated to be around 1.28 s. As is shown in Fig. 1, for a typical beam and column section, the column longitudinal reinforcements were distributed around the section, while the beam longitudinal bars were positioned at the top and bottom of the section in all frames. Fig. 2 provides a schematic illustration along with the dimensions and flexural/shear reinforcement of the members in the considered frames. The deformed steel bar, 10 mm in diameter, was utilised as shear reinforcement.

## 4. Pushover analysis of the original frame

Due to the inelastic behaviour of structures under severe ground motions, recourse to nonlinear analysis is unavoidable when it comes to the seismic assessment of structures. Generally, nonlinear procedures comprise of nonlinear static (pushover) analysis and nonlinear dynamic (time history) analysis. In pushover analysis, a monotonically increasing lateral load in addition to a constant gravity load is applied to the structure to a predefined lateral displacement or to the onset of plastic collapse mechanism.

By providing elaborate numerical modelling, precise definition of nonlinear properties of structural components, and selection of a suitable set of ground motion records, the nonlinear time history analysis would result in a more accurate prediction of the seismic response of a structure. However, within the boundary of its limitation, pushover analysis could provide a good estimate of seismic response [22–24] in lieu of a time history analysis.

In this section, the results of pushover analysis of the unreinforced original frame are presented. The inelastic behaviour of each beam and column member has been allowed using the lumped plasticity concept. After verification of the adopted assumptions and results of the nonlinear analysis, a pushover analysis of the frame was carried out in SAP 2000 [16]. The base shear versus roof displacement curve was used as a characteristic force–displacement curve.

### 4.1. Determination of nonlinear hinge properties

Nonlinear analysis of RC structures using concentrated plastic hinges requires the determination of inelastic properties of each

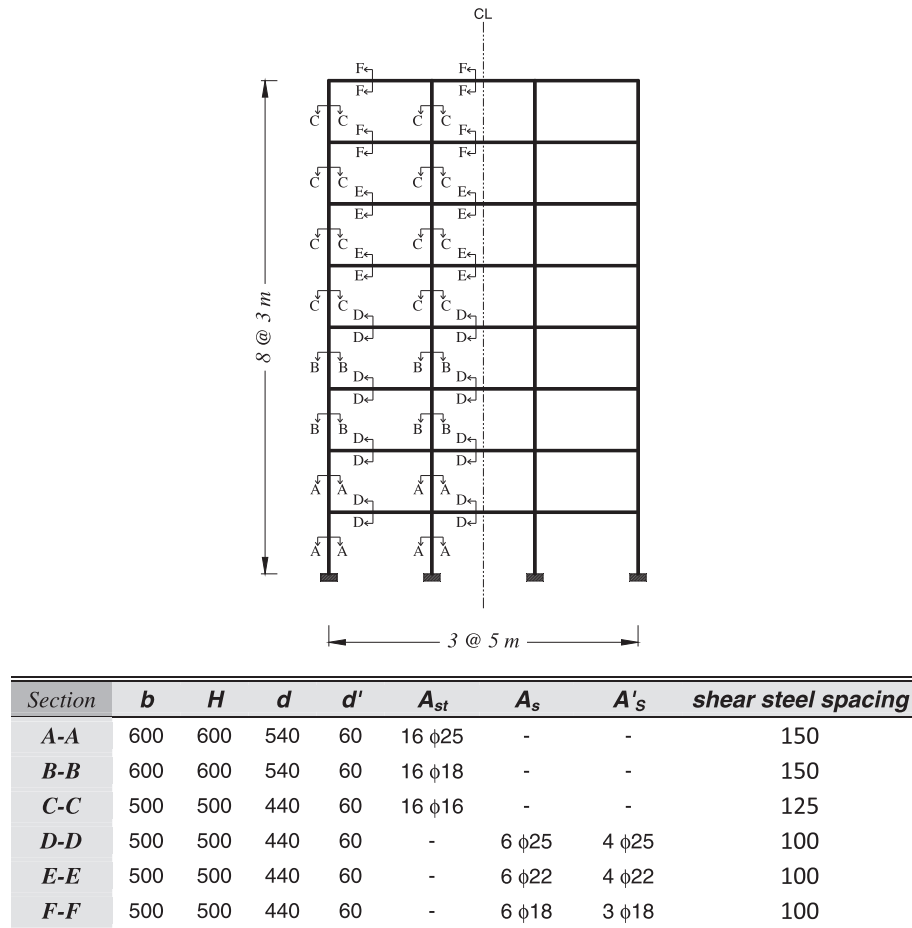


Fig. 2. Reinforcement details of the 8-storey intermediate frame (all dimensions are in mm).

structural member quantified by force and displacement. In the lumped plasticity approach, a rigid plastic hinge could be defined for every degree of freedom. For a comprehensive nonlinear analysis, consideration of the shear failure in members is necessary. However, some past studies [14,25,26] have reported that even for under-designed RC buildings, possessing inadequate transverse reinforcement, the shear demand is significantly lower than the shear capacity in both beams and columns and that no shear failure would occur. Therefore, in this study, only inelastic flexural behaviour of structural elements was considered through concentrated rigid plastic hinges at the ends which are susceptible to inelastic behaviour.

The moment–curvature relationships of the end sections in the beams and columns were determined based on fibre analysis using XTRACT software [27], considering section properties, reinforcement details and a constant axial load. Axial loads on the columns were assumed to be equal to the resultant loads calculated from dead loads plus 20% of live loads (as recommended in the Iranian seismic code [19]). On the beams, the axial forces due to gravity loads were assumed to be equal to zero. The commonly used confined concrete model proposed by Mander et al. [28], was implemented while an elastic perfectly plastic model with parabolic strain hardening was considered for steel. The inelastic properties recommended in ASTM A615M [29] were used for the steel reinforcement.

The real moment–curvature (or moment–rotation) curve of a RC member in which the tension steel yields, can be idealised to a simplified bilinear curve, as shown in Fig. 4 for a typical RC beam [30]. In Fig. 3, point B corresponds to the tensile yield strain in

the steel indicating yield moment,  $M_y$ , and yield rotation,  $\theta_y$ , while point C corresponds to the ultimate conditions; namely ultimate moment,  $M_u$ , and corresponding ultimate rotation,  $\theta_u$ . The ultimate condition was considered to be the attainment of one of the following conditions; whichever happened first [30].

- (1) A 20% drop in the moment capacity of member.
- (2) When the tensile strain in the longitudinal steel reaches the ultimate tensile strain.
- (3) The attainment of the ultimate compression strain in concrete using the following equation proposed by Scott et al. [31].

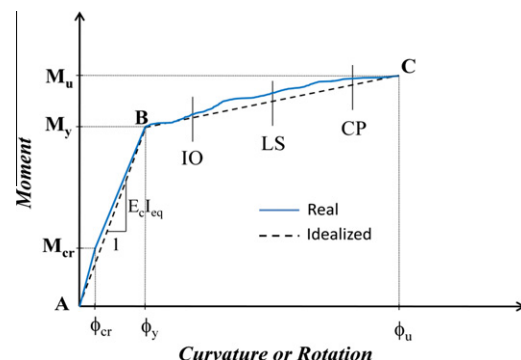


Fig. 3. A typical real moment–rotation (or moment–curvature) and the corresponding idealised curve for a flexural plastic hinge.

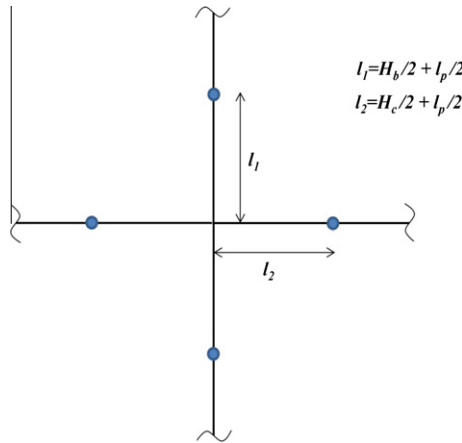


Fig. 4. Hinge locations at the beams and columns for a typical interior joint.

$$\varepsilon_{cu} = 0.004 + 0.9\rho_s \left( \frac{f_{yh}}{300} \right) \quad (1)$$

In the above equation,  $\varepsilon_{cu}$  is the ultimate compressive strain of concrete,  $\rho_s$  is the volumetric ratio of confining reinforcement and  $f_{yh}$  represents the yield strength of transverse steel.

Plastic rotation is defined as the difference between the ultimate and the yield curvature (curvature ductility) multiplied by the plastic hinge length. Many equations have been proposed by researchers for the plastic hinge length [30,32]. In this study, the plastic hinge was calculated by:

$$l_p = H/2 \quad (2)$$

where  $L_p$  and  $H$  are the plastic hinge length and the height of section, respectively. The adopted plastic hinge length is also recommended by ATC-40 and by other researchers [33]. It should be noted that according to Paulay and Priestley [32], Eq. (2) results in accurate values for the conventional beam and column sections. Although not the main focus of this study, the acceptance criteria of immediate occupancy (IO), life safety (LS) and collapse prevention (CP) were defined for the beam and columns similar to the ratios recommended in FEMA-356 [21].

The flexural and axial-flexural hinges were introduced at both ends of beams and columns, respectively. For the columns, the yield moment changes according to the axial load. Thus, a yield moment-axial load interaction curve needs to be defined for each column. The hinges were located considering the beam and column dimensions and the plastic hinge length, as shown in Fig. 4 where  $H_c$  and  $H_b$  stand for the depth of columns and beams, respectively [25]. The simplified assumption of shear span,  $L_v = 0.5L$  was adopted during the lateral loading of the structure in which  $L$  refers to the total member length [14,30].

#### 4.2. Verification of the nonlinear results

The assumptions and models adopted in the last section in order to quantify the plastic hinge properties was verified against the pushover results of a ductile RC frame reported by Filiatrault et al. [34,35]. In their study, the seismic performance of the frame was investigated using shake table tests. Their results also were confirmed in a nonlinear analysis based on the concepts of spread plasticity. The test structures were designed, at a reduced scale, according to the provisions of the National Building Code of Canada [36] and the Canadian concrete standard [37]. The structure was assumed to be part of the lateral load resisting system of a building, with two stories (each 1.5 m high) and two bays (each 2.5 m wide). The ductile frame incorporated full seismic details compos-

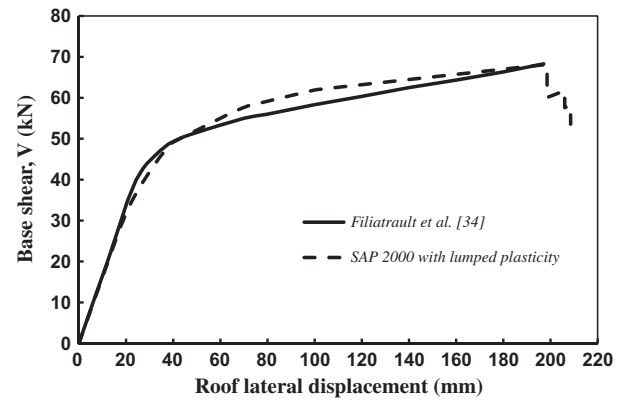


Fig. 5. Comparison of pushover curves of ductile frame.

ing of rectangular hoops, with 135° hooks, spaced at 30 mm in the centre at the critical locations of the beams, columns, and joints.

In building the numerical model in SAP 2000, the lateral load distribution and applied gravity loads were identical to those used by Filiatrault et al. [34]. Determination of the plastic hinge properties of each structural member was based on the assumed models for concrete and steel described in the current study. However, the actual material properties obtained from tensile test on reinforcing steel and compressive test on concrete cylinders were considered [35].

Fig. 5 compares the base shear versus roof lateral displacement curves extracted from the nonlinear pushover analysis of the present study with that obtained by Filiatrault et al. [34]. Despite some discrepancy, the two load-displacement curves particularly in terms of the failure point agree well, indicating the reliability of the adopted assumptions for the characterisation of the plastic hinge properties. In addition, the sequence of plastic hinging in nonlinear analysis using the adopted lumped plasticity conforms to the capacity design philosophy adopted in the Canadian concrete standard [37]. The first four hinges occurred in the beams followed by three others at the base of the columns without any inelastic behaviour at the top of the columns up to the point of failure. This hinging pattern was similar to that found by Filiatrault et al. [34,35] both experimentally and numerically.

#### 4.3. Pushover results of the original frame

Once the analysis results were validated, nonlinear pushover analysis of the frame was performed to obtain the total resistant and lateral displacement capacity of the structure. In order to provide an accurate seismic response of a structure, nonlinear analysis should include the effect of probable gravity load during an earthquake. In this study, the total dead load plus 20% of the live load based on the Iranian seismic code [19], is considered during the pushover analysis. In seismic evaluation of a building, the lateral force profile should represent, though approximately, the likely distribution of inertial forces induced during an earthquake. In the current paper, an inverted triangular distribution over the height is used as the lateral load pattern. According to a comparative study conducted by Mwafy and Elnashai [24], this pattern provides better estimates of the capacity curve and seismic responses in comparison to a uniform distribution. In addition, while inverted triangular distribution is more practical than multi-modal distribution, it would yield similar results. It should be mentioned that the selected load pattern is similar to the lateral load distribution used for the seismic design of considered structures and has been suggested in the Iranian seismic code. Also, the effects of the second moments attributed to the axial loads in the deformed members ( $P - \Delta$  effects) have been considered in the nonlinear analysis.

**Table 1**  
Stiffness properties for nonlinear static analysis of the frames.

| Member type            | Section | Story no. | Pos.     | $E_c I_{eq}$ (kN m <sup>2</sup> ) | $I_{eq}/I_g$ |
|------------------------|---------|-----------|----------|-----------------------------------|--------------|
| Columns                | A–A     | 1         | Interior | 1.27 E5                           | 0.50         |
|                        |         |           | Exterior | 1.21 E5                           | 0.48         |
|                        |         | 2         | Interior | 1.26 E5                           | 0.49         |
|                        |         |           | Exterior | 1.19 E5                           | 0.47         |
|                        | B–B     | 3         | Interior | 9.22 E4                           | 0.36         |
|                        |         |           | Exterior | 8.27 E4                           | 0.33         |
|                        |         | 4         | Interior | 8.89 E4                           | 0.35         |
|                        |         |           | Exterior | 8.01 E4                           | 0.32         |
|                        | C–C     | 5         | Interior | 4.45 E4                           | 0.36         |
|                        |         |           | Exterior | 4.06 E4                           | 0.33         |
|                        |         | 6         | Interior | 4.25 E4                           | 0.35         |
|                        |         |           | Exterior | 3.89 E4                           | 0.32         |
|                        |         | 7         | Interior | 4.06 E4                           | 0.33         |
|                        |         |           | Exterior | 3.72 E4                           | 0.3          |
|                        |         | 8         | Interior | 3.72 E4                           | 0.3          |
|                        |         |           | Exterior | 3.54 E4                           | 0.29         |
| Mean value for columns |         |           |          |                                   | 0.37         |
| Beams                  | D–D     | 1–4       |          | 6.22 E4                           | 0.51         |
|                        | E–E     | 5–6       |          | 5.18 E4                           | 0.42         |
|                        | F–F     | 7–8       |          | 3.81 E4                           | 0.31         |
| Mean value for beams   |         |           |          |                                   | 0.41         |

Due to the flexural cracking of RC members, stiffness of the members is reduced during seismic loading. Reduction in the flexural stiffness was considered in the nonlinear analysis by introducing the equivalent moment of inertia,  $I_{eq}$ , given by:

$$I_{eq} = \frac{M_y}{E_c \phi_y} \quad (3)$$

where  $M_y$  and  $\phi_y$  are the yield moment and the yield curvature, respectively; and  $E_c$  is the elastic modulus of concrete which was taken as [17].

$$E_c = 4700 \sqrt{f'_c} \quad (4)$$

in which  $f'_c$  is the compressive strength of concrete.

The stiffness properties of all members along with the ratios of the equivalent moment of inertia ( $I_{eq}$ ) to the gross moment of inertia ( $I_g$ ) are summarised in Table 1.

Despite the intermediate design provisions of ACI 318-02 did not incorporate the weak-beam strong-column design philosophy; nonlinear results of the frame indicated a beam sidesway mechanism as shown in Fig. 6. This could be particularly on account of complying with the maximum inter-story drift as suggested in the Iranian seismic code. However, columns of the mid-stories suffered from severe inelastic behaviour. In Fig. 7, the total lateral load was plotted against the roof lateral displacement.

## 5. Design of retrofitting with composites

With the aim of increasing the total lateral resistance of the structure, it was decided to flexurally strengthen structural members at regions prone of inelastic behaviour. Fig. 8 shows a schematic illustration of retrofitting configuration at a typical interior joint. The efficiency of this FRP retrofitting method in relocating plastic hinges away from the column face towards the beam in beam–column sub-assemblages has been proved previously [3]. However, the capability of FRP retrofits to relocate nonlinear hinges is heavily dependent on the length of composite sheets. If this length exceeds a threshold (usually around the as beam depth), nonlinear plastic hinges would form at the location similar to the original beam–column joint, but with higher yield strength. In the current study, the length of composite sheets in beams and

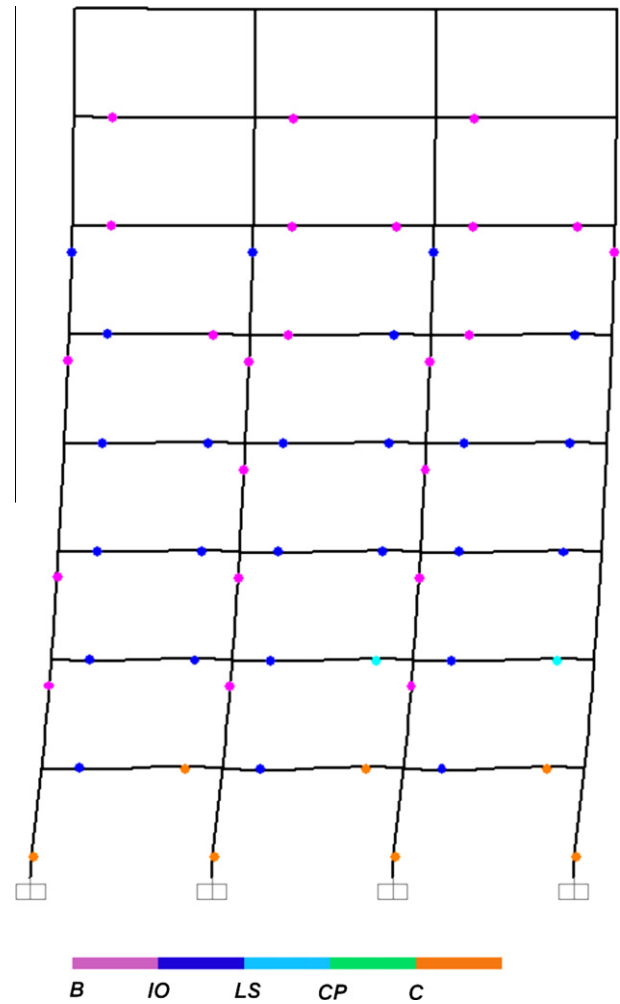


Fig. 6. Hinging pattern and damage level of the original frame.

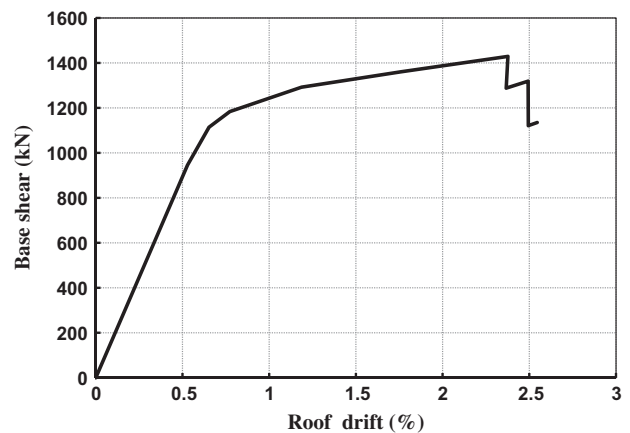


Fig. 7. Pushover curve of the original frame.

columns was assumed to be long enough to eliminate the likelihood of plastic hinge relocation.

Debonding of composite sheets from the concrete substrate is a major concern that affects the application of externally bonded FRP in RC members, especially in flexural strengthening. This issue may precipitate the premature failure of the retrofitted member [38]. To overcome this problem, a number of methods have been suggested in recent years. In an experimental research, Mostofinejad and



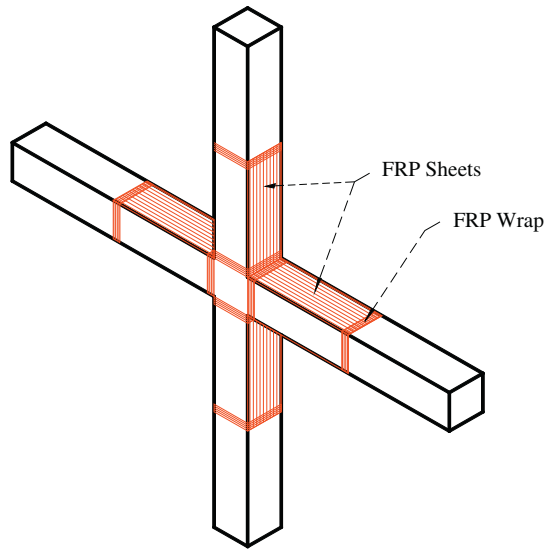


Fig. 8. Schematic illustration of FRP retrofit configuration in an interior joint.

Mahmoudabadi [39] developed an innovative grooving technique. Their experimental results indicated that by providing adequate depth, longitudinal grooves would completely preclude debonding. Other methods such as wrapping of composite sheets with adequate thickness and FRP anchors were also proved to be capable of efficiently eliminate the problem [15,40–42]. In another experimental study, Orton et al. [41] concluded that bonding FRP sheets using adequate number of CFRP anchors with effective embedment depth allows the development of the full tensile capacity of CFRP sheets. Taking the above into consideration, it was assumed that debonding can be prevented by appropriate application of FRP anchors and wraps. Detailed design of these anchors is out of the scope of this study. The design of such FRP anchors can be found elsewhere [41,42].

In order to compare the efficiency of GFRP and CFRP composites in flexural strengthening of RC structures, it was decided to apply both composite sheets. Seismic response of GFRP and CFRP retrofitted frames were compared with the original frame in terms of lateral force–displacement curve, hinging pattern, and inter-story drift. Design properties of unidirectional GFRP and CFRP compos-

ites were identical to those adopted in De Luca et al. [43] and Mahini and Ronagh [2], respectively (see Table 2).

### 5.1. GFRP retrofitted frame

In order to create a less ductile failure of the retrofitted member, it is desired that the ultimate concrete strain is attained prior to the FRP rupture [44]. The initial moment–curvature analysis of the beams proved that two layers of GFRP sheets were needed to be externally bonded on the top and bottom sides of the beams.

According to the capacity design concept adopted in modern seismic design, flexural capacity of the columns at each connection shall be higher than the summation of beams' flexural strength to satisfy the weak-beam strong-column design philosophy. Therefore, the thickness of composite sheets in the columns was assumed to be identical to the beams, except the columns of fifth and sixth stories (section C–C) where four layers of composite sheets were calculated to be needed to comply with the capacity design concept. It should be mentioned that the pushover analysis of the GFRP retrofitted frame using two layers of composite sheets in all members showed a column sidesway mechanism due to the failure of plastic hinges at the columns of the fifth story.

Generally, flexural strengthening of RC members using FRP leads to a considerable loss of ductility. To provide the retrofitted structures with adequate energy dissipation capacity, the retrofitting design strategy was aimed at augmenting the lateral resistance without substantial loss of lateral displacement capacity and significant differences in the inter-story drift values. To pursue this objective, it was found that strengthening of beams at the top stories could significantly affect the displacement capacity and inter-story drifts undertaken by the retrofitted frame. After a series of sensitivity analysis and to provide an even inter-story drift distribution, the beams in the last three stories were left un-retrofitted. Table 3 summarises the number of GFRP sheets that were applied on the flange sides of each member. It should be mentioned that if more lateral displacement is desired, it could be achieved with FRP wrapping of RC columns.

### 5.2. CFRP retrofitted frame

To provide an equal condition for comparison purposes, the thickness of composite sheets in the beams of CFRP retrofitted frame was selected to generate the ultimate tensile strength equal

Table 2  
Mechanical properties of composite sheets.

| Composite type | Tensile strength $f_f$ (MPa) | Ultimate tensile strain, $\epsilon_{ff}$ | Tensile modulus $E_f$ (MPa) | Thickness $t_f$ (mm) |
|----------------|------------------------------|------------------------------------------|-----------------------------|----------------------|
| CFRP           | 3900                         | 0.0155                                   | 240,000                     | 0.165                |
| GFRP           | 3241                         | 0.045                                    | 72,397                      | 0.589                |

Table 3  
Number of composite sheets in the GFRP retrofitted frame.

| Section             | A–A     | B–B     | C–C     |         | D–D | E–E | F–F |
|---------------------|---------|---------|---------|---------|-----|-----|-----|
| Story no.           | 1 and 2 | 3 and 4 | 5 and 6 | 7 and 8 | 1–4 | 5   | 6   |
| No. of GGFRP layers | 2       | 2       | 4       | 4       | 2   | 2   | –   |

Table 4  
Number of composite sheets in the CFRP retrofitted frame.

| Section            | A–A     | B–B     | C–C     |         | D–D | E–E     | F–F     |
|--------------------|---------|---------|---------|---------|-----|---------|---------|
| Story no.          | 1 and 2 | 3 and 4 | 5 and 6 | 7 and 8 | 1–4 | 5 and 6 | 7 and 8 |
| No. of CFRP layers | 6       | 6       | 9       | 6       | 6   | 6       | –       |

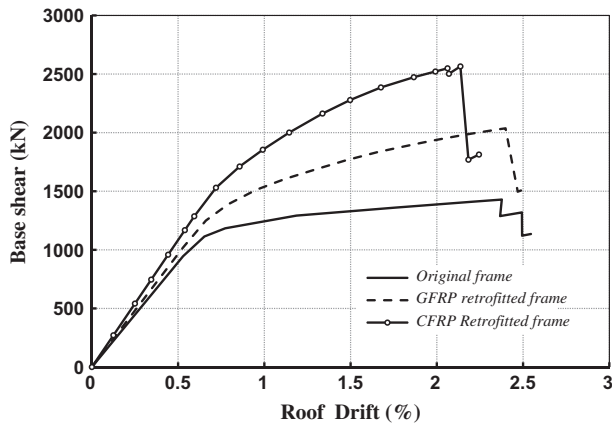


Fig. 9. Comparison of the pushover curves of the retrofitted and original frames.

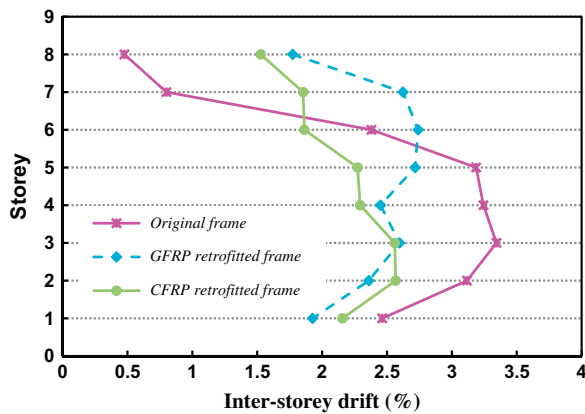


Fig. 10. Inter-storey drift distribution of original and retrofitted frames.

to the GFRP retrofitted frame. As a result and taking the mechanical properties of GFRP and CFRP sheets into consideration, six layers of CFRP was selected for the beams in all stories except the two last beams (section F–F in Fig. 2). This strengthening configuration guaranteed an approximately identical inter-story drift at all stories while preserving adequate lateral displacement capacity for the frame. In order to respect the weak-beam strong column design philosophy in the retrofitted frame, a similar amount of composite was applied to the tensile sides of the columns at all stories. However, similar to the GFRP retrofitted frame, classical calculation and primary nonlinear pushover analysis confirmed column hinging at the fifth story. To eliminate this undesirable performance, nine CFRP layers were found to be required in order to satisfy the weak-beam strong-column condition in the columns of fifth and sixth stories (section C–C). The final designated amounts of composite sheets are tabulated in Table 4. Nonlinear results also confirmed that by providing confined concrete at the columns, more lateral displacement capacity would be achieved.

## 6. Discussion of the analysis results of the retrofitted frames

Nonlinear static analysis of the frames retrofitted based on the aforementioned configuration indicated a significant increase in the lateral resistance. Of particular interest was satisfying the weak-beam strong-column design principle in the design of retrofitting configuration. As shown in Fig. 9, the load carrying capacity of the structure could be increased by 43% and 80%, using GFRP and CFRP intervention, respectively. However, while the former provided a lateral displacement capacity almost identical to the original frame, the latter decreased it by around 10% which is not significant compared to the acquired strength increment. It should be mentioned that the lateral displacement of the CFRP retrofitted frame would be increased to be identical to the original frame if the beams in the three last stories of two were not retrofitted. However, this configuration may lead to the plastic hinge forma-

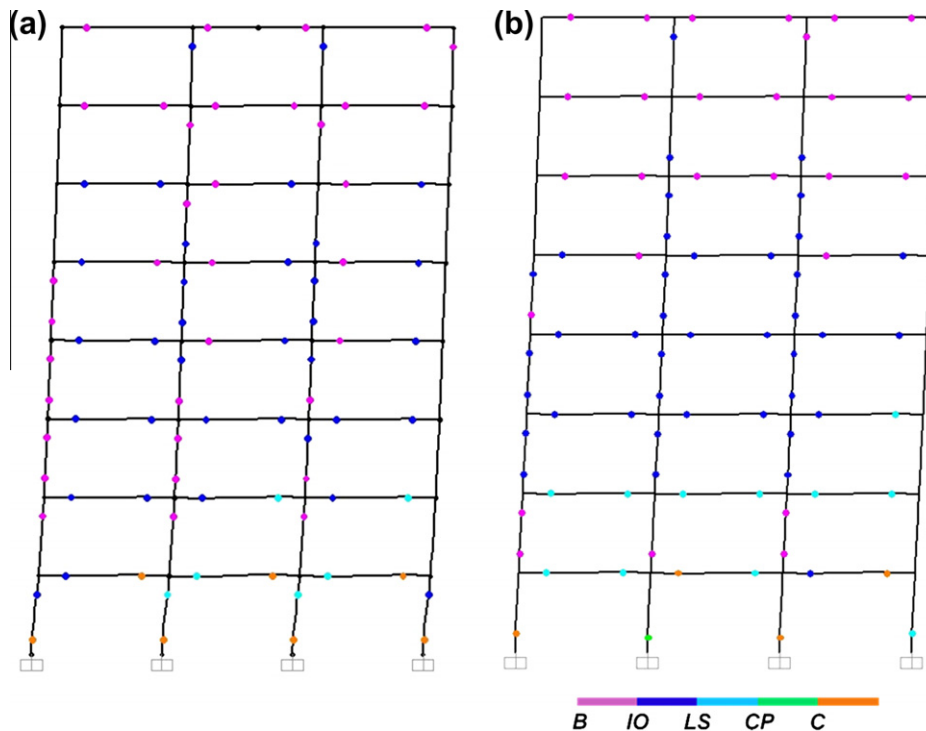


Fig. 11. Plastic hinge pattern of: (a) GFRP (b) CFRP retrofitted frame at ultimate load.

tion at the columns of the sixth story which could not follow the capacity design concepts adopted in modern seismic design.

Although the section analysis of structural members confirmed a significant decrease in the curvature ductility, especially for CFRP retrofitted sections, the pushover curves exhibited a negligible effect in the displacement capacity. It should be mentioned that although different FRP configuration would alter the displacement capacity of the frame, the CFRP retrofitted frame found to possess a higher load carrying capacity compared to the GFRP retrofitted frame.

The selection of retrofit design configurations were based on the assumption of a uniform distribution of inter-story drift for all stories. In a regular structure which is the case of the selected frame, this could preclude force and deformation concentration in one particular story during strong ground motion. As observed in Fig. 10, the lateral inter-story drifts of different stories in the rehabilitated frames are almost similar, varying from 1.5% to 2.5%; this is different to the ones observed for the original frame from which were from 0.5% to 3.5%. Comparing the inter-story drift in GFRP and CFRP retrofitted frames; it is seen that the former suffers from higher values in the upper stories in spite of providing more composite sheets at the columns of fifth and sixth stories. This might be particularly due to higher strength of CFRP retrofitted columns compared to the GFRP retrofitted ones. It is worth mentioning that the complete collapse of a structure according to SEAOC (Vision 2000) [45], is assumed once the inter-story drift exceeds 2.5%. Implementation of this global collapse criterion underlines much lower lateral resistance in the original frame emphasising the efficiency of composite materials, especially CFRP in extracting the whole lateral resistance of the structure.

The hinging pattern of the rehabilitated frames conform a desired beam sidesway failure mechanism as illustrated in Fig. 11. Similar to the original frame, the order of hinging development is according to the weak-beam strong-column design philosophy. Compared to the original frame, more beams at the top stories of the rehabilitated structures sustained inelastic behaviour. It could be concluded that a well-designed strengthening configuration using composite materials, could extract all member's strength providing more lateral strength for the structure.

## 7. Summary and conclusions

This paper reports on the results of a numerical investigation into the seismic response of a code-compliant RC structure rehabilitated using composite sheets. To provide a comparative basis, two typically used composite materials (GFRP and CFRP) were externally bonded to the tensile sides of the structural members in the inelastic prone regions. This configuration is practical in real applications. Without undermining the lateral displacement capacity, the retrofitting design strategy was aimed at increasing the lateral resistance of the structure in which inter-story drift values were distributed uniformly.

The nonlinear pushover analysis of the frames outlined a dramatic improvement of the lateral load carrying capacity without significant loss of displacement ductility. Moreover, it was found that retrofitting configuration could affect the seismic response of the structures substantially. Based on the decided rehabilitation strategy, GFRP and CFRP retrofitting could increase the lateral resistance of the structure about 43% and 80%, respectively. Neglecting the slight decrease of lateral displacement using CFRP, it could be concluded that in terms of improving the lateral resistance of the structures, carbon fibre composites are more efficient than glass fibre composites. In addition, the hinging patterns of the retrofitted frames conform a desirable beam sidesway condition.

## References

- [1] Le-Trung K, Lee K, Lee J, Lee DH, Woo S. Experimental study of RC beam-column joints strengthened using CFRP composites. *Compos Part B: Eng* 2010;41:76–85.
- [2] Mahini SS, Ronagh HR. Web-bonded FRPs for relocation of plastic hinges away from the column face in exterior RC joints. *Compos Struct* 2011;93:2460–72.
- [3] Dalalbashi A, Eslami A, Ronagh HR. Plastic hinge relocation in RC joints as an alternative method of retrofitting using FRP. *Compos Struct* 2012;94:2433–9.
- [4] Ha G-J, Cho C-G, Kang H-W, Feo L. Seismic improvement of RC beam-column joints using hexagonal CFRP bars combined with CFRP sheets. *Compos Struct*. doi.org/10.1016/j.compstruct.2012.08.022.
- [5] Attari N, Amziane S, Chemrouk M. Efficiency of beam-column joint strengthened by FRP laminates. *Adv Compos Mater* 2010;19:171–83.
- [6] Ascione L, Berardi VP, Feo L, Mancusi G. A numerical evaluation of the interlaminar stress state in externally FRP plated RC beams. *Compos Part B: Eng* 2005;36:83–90.
- [7] D'Ambrisi A, Feo L, Focacci F. Bond-slip relations for PBO-FRCM materials externally bonded to concrete. *Compos Part B: Eng*. 2012;43:2938–49.
- [8] D'Ambrisi A, Feo L, Focacci F. Experimental analysis on bond between PBO-FRCM strengthening materials and concrete. *Compos Part B: Eng*. 2013;44:524–32.
- [9] Seo S-Y, Feo L, Hui D. Bond strength of near surface-mounted FRP plate for retrofit of concrete structures. *Compos Struct*. <http://dx.doi.org/10.1016/j.compstruct.2012.08.038>.
- [10] Ascione L, Feo L. Modeling of composite/concrete interface of RC beams strengthened with composite laminates. *Compos Part B: Eng* 2000;31:535–40.
- [11] Yuan H, Lu X, Hui D, Feo L. Studies on FRP-concrete interface with hardening and softening bond-slip law. *Compos Struct* 2012;94:3781–92.
- [12] Ascione F, Berardi VP, Feo L, Giordano A. An experimental study on the long-term behavior of CFRP pultruded laminates suitable to concrete structures rehabilitation. *Compos Part B: Eng* 2008;39:1147–50.
- [13] Balsamo A, Colombo A, Manfredi G, Negro P, Protà A. Seismic behavior of a full-scale RC frame repaired using CFRP laminates. *Eng Struct* 2005;27:769–80.
- [14] Di Ludovico M, Protà A, Manfredi G, Cosenza E. Seismic strengthening of an under-designed RC structure with FRP. *Earthquake Eng Struct Dynam* 2008;37:141–62.
- [15] Garcia R, Hajirasouliha I, Pilakoutas K. Seismic behaviour of deficient RC frames strengthened with CFRP composites. *Eng Struct* 2010;32:3075–85.
- [16] Computers and structures Inc. Static and dynamic finite element analysis of structures. SAP 2000. 14.1.0 ed. Berkeley (CA); 2009.
- [17] ACI committee 318. Building code requirements for structural concrete (ACI 318-02) and commentary (ACI 318R-02). Farmington Hills (MI): American Concrete Institute; 2002.
- [18] ACI committee 318. Building code requirements for structural concrete (ACI 318-08) and commentary (ACI 318R-08). Farmington Hills (MI): American Concrete Institute; 2008.
- [19] Permanent committee for revising the Iranian code for seismic resistant design of buildings. Iranian code of practice for seismic resistant design of buildings (Standard No. 2800-05). 3rd ed. Tehran (Iran): Building and Housing Research Center (BHRC); 2005.
- [20] International conference of building officials. Uniform Building Code (UBC). Whittier (CA); 1994.
- [21] American society of civil engineering (ASCE). Prestandard and commentary for the seismic rehabilitation of buildings (FEMA-356). Washington (DC): Federal Emergency Management Agency; 2000.
- [22] Krawinkler H, Seneviratna GDPK. Pros and cons of a pushover analysis of seismic performance evaluation. *Eng Struct* 1998;20:452–64.
- [23] Sadjadi R, Kianoush MR, Talebi S. Seismic performance of reinforced concrete moment resisting frames. *Eng Struct* 2007;29:2365–80.
- [24] Mwafy AM, Elnashai AS. Static pushover versus dynamic collapse analysis of RC buildings. *Eng Struct* 2001;23:407–24.
- [25] Inel M, Ozmen HB. Effects of plastic hinge properties in nonlinear analysis of reinforced concrete buildings. *Eng Struct* 2006;28:1494–502.
- [26] Jeong SH, Elnashai AS. Analytical assessment of an irregular RC frame for full-scale 3D pseudo-dynamic testing Part I: Analytical model verification. *J Earthquake Eng* 2005;9:95–128.
- [27] Imbeon and Associates Inc. Cross section analysis program for structural engineers. XTRACT. 3.0.8 ed. California; 2011.
- [28] Mander JB, Priestley MJN, Park R. Theoretical stress-strain model for confined concrete. *J Struct Eng* 1988;114:1804–26.
- [29] ASTM A615M. Standard specification for deformed and plain carbon-steel bars for concrete reinforcement. Philadelphia (PA): American Society for Testing and Materials; 2009.
- [30] Park R, Paulay T. Reinforced concrete structures. New York: Wiley; 1975.
- [31] Scott BD, Park R, Priestley MJN. Stress-strain behavior of concrete confined by overlapping hoops at low and high strain rates. *J Am Concr Inst* 1982;79:13–27.
- [32] Paulay T, Priestley MJN. Seismic design of reinforced concrete and masonry buildings. New York: Wiley; 1992.
- [33] Zou XK, Teng JG, De Lorenzis L, Xia SH. Optimal performance-based design of FRP jackets for seismic retrofit of reinforced concrete frames. *Compos Part B: Eng* 2007;38:584–97.



- [34] Filiatrault A, Lachapelle E, Lamontagne P. Seismic performance of ductile and nominally ductile reinforced concrete moment resisting frames. II. Analytical study. *Can J Civil Eng* 1998;25:342–52.
- [35] Filiatrault A, Lachapelle E, Lamontagne P. Seismic performance of ductile and nominally ductile reinforced concrete moment resisting frames. I. Experimental study. *Can J Civil Eng* 1998;25:331–41.
- [36] NBCC. National building code of Canada. Ottawa, Ont.: Associate Committee on the National Building Code, National Research Council of Canada; 1995.
- [37] CSA. Design of concrete structures for building. Standard CAN-A-233-94. Rexdale (Ont.): Canadian Standards Association; 1994.
- [38] Oehlers DJ. Development of design rules for retrofitting by adhesive bonding or bolting either FRP or steel plates to RC beams or slabs in bridges and buildings. *Compos – Part A: Appl Sci Manuf* 2001;32:1345–55.
- [39] Mostofinejad D, Mahmoudabadi E. Grooving as alternative method of surface preparation to postpone debonding of FRP laminates in concrete beams. *J Compos Constr* 2010;14:804–11.
- [40] Ozdemir G. Mechanical properties of CFRP anchorages. MSc thesis. Department of Civil Engineering, Turkey: Middle East Technical University; 2005.
- [41] Orton SL, Jirsa JO, Bayrak O. Design considerations of carbon fiber anchors. *J Compos Constr* 2008;12:608–16.
- [42] Zhang HW, Smith ST, Kim SJ. Optimisation of carbon and glass FRP anchor design. *Constr Build Mater* 2012;32:1–12.
- [43] De Luca A, Nardone F, Matta F, Nanni A, Lignola GP, Prota A. Structural evaluation of full-scale FRP-confined reinforced concrete columns. *J Compos Constr* 2011;15:112–23.
- [44] El-Mihilmy MT, Tedesco JW. Analysis of reinforced concrete beams strengthened with FRP laminates. *J Struct Eng* 2000;126:684–91.
- [45] SEAOC. Performance based seismic engineering of buildings. Vision 2000 committee. Sacramento (CA): Structural Engineers Association of California; 1995.